

Riddhiman Raguraman

Graphics Programmer

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🔗 Riddhiman Raguraman 📁 Portfolio Website

Currently pursuing a Master's degree in Game Programming. Authorized to work in the United States under F-1 Optional Practical Training (OPT), with STEM extension eligibility.

Education

MS in Game Programming, Depaul University [🔗](#)

09/2024 - present | Chicago, Illinois

2.92 in 4 point CGPA

Btech Computer Science, VIT Chennai [🔗](#)

2020 - 2024 | Chennai, India

8.10 in 10 point CGPA | 3.24 in 4 point CGPA

12th grade, Velammal Vidyalaya [🔗](#)

2020 | Chennai, India

91%

Languages

Tamil – Native Language | English – IELTS 7

Projects

Game Development Portfolio [🔗](#)

Website made to showcase my Graphics Programming, Game Development Projects and others

Game Engine Development, (C++/DirectX11)

- **Math System**: - Programmed fully-fledged vector, matrix and quaternion math libraries. - Improved libraries performance by a factor of 4 by using SIMD Intrinsics (MMX and SSE).
- **File System**: - Created read/write data storing mechanism using OS native functions for improved performance. - Developed archiving system for packing/retrieving different data structures into single files.
- **Object System**: - Implemented a tree-based data structure for handling hierarchies of objects in memory. - Employed a system for implementing scene graphs and an animation system in the engine.
- **Graphics System**: - Developed **Directx-based** graphics engine using vertex buffers and vertex array objects. - System capable of frustum culling, multiple cameras, object transformations, and mesh loading.
- **Converter System**: - Created an offline conversion tool for converting mesh data from GLTF files into a runtime file format based on Google Protocol buffer.

Custom Malloc and Free [🔗](#), (Windows C++ console project)

Scope: Developed to manage memory allocation and deallocation efficiently in performance-critical applications, utilizing a custom memory management system. My custom Malloc and Free are 2 times faster and MSVC's **Created For**: Optimised C++ course **Key Technologies**: designing and implementing an optimized memory allocation framework, focusing on C++ integration and system-level memory management.

Multi threaded Audio engine, (C++ Multithreading Project)

Scope : Engineered a **high-performance audio playback system** to seamlessly stitch and play 23 raw WAVE files in real-time without artifacts. The architecture orchestrates 25 concurrent threads to decouple file I/O from audio rendering, ensuring glitch-free playback. **Created For** : Advanced C++ Concurrency Course **Key Technologies** : implementing complex thread synchronization using mutex , condition_variable , and unique_lock ; utilizing a **double-buffered Producer-Consumer** pattern in the Coordinator thread; employing circular queues for efficient inter-thread messaging; ensuring thread safety and resource management via RAII; optimizing performance to meet strict real-time audio constraints without spinlocks.

Professional Experience

Game Developer Intern, Geonix Logix

05/2023 - 09/2023

Worked on a 2D educational game called 'Clouds Rescue.' This game was designed to make learning math fun and interactive for 10th-grade students. Was responsible for developing the entire gameplay, handling level design, and writing all the interaction code.

Courses

Master Certificate in AR/VR [🔗](#),

Monolith Research and Training labs

Operating Systems and You: Becoming a Power User [🔗](#)

Data Science [🔗](#)

Multi threaded Maze solver, (C++ multithreading Project)

Scope: Engineered a **high-performance multithreaded maze solver** capable of navigating massive 20Kx20K grids using a bidirectional search strategy. The solution launches concurrent threads from both start and end points to converge on a solution, significantly outperforming traditional single-threaded BFS/DFS algorithms in execution time by 15 to up to of 50% . **Created For:** Optimized C++ Multithreading Course **Key Technologies:** implementing lock-free synchronization using **C++ Atomics** to manage shared state without mutex overhead, and **utilizing bitwise operations** for efficient data processing.

Cloud's Rescue , (Unity 2d Project)

Scope: Developed as an educational 2D mobile game tailored for 10th-grade students, Clouds Rescue combines an engaging storyline with challenging math-based levels to make learning mathematics an enjoyable experience. **Created For:** Internship **Role:** implementing game mechanics using C#, integrating characters crafted with Blender, and creating dynamic gameplay that fostered critical thinking and problem-solving. **Key Technologies:** Unity 2D, C#, ShaderLab and Blender.

Skills

C++ – Expert

I have 8 years of Knowledge in C++, extending beyond fundamental programming into advanced areas. This includes proficiency with the **STL** libraries, **design patterns**, **algorithms**, and expertise in Intel intrinsic programming (**SIMD**), and **various optimization techniques**. I also have foundational skills in multi-threading programming and working with graphics libraries.

Unreal Game Engine

I've been using Unreal Engine extensively for the past year, gaining expertise in **Blueprints Coding**, **Level Design**, **Materials Creation**, **Animation**, **Sound Design**, **Architectural Visualization**, **VFX**, and **VR**.

Other Computer Languages

I also have a working knowledge of C, C#, Java, Python, Assembly, and Bash.

Graphics Frameworks – Proficient

I have been working with **DirectX 11** and shader languages like **HLSL**. I have knowledge of the rendering pipeline, extending beyond just **compute shaders** to **include context switching, aliasing, texture mapping**, and more. I also have working knowledge of **Vulkan**.

Unity Game Engine

Have been using Unity for the past 4 years, and have extreme knowledge of **asset management**, and **coding in C#**, and enough knowledge in animation and deployment of the game,

Platforms, Version control and Scripting

Know to handle **Windows and Linux OS**, and scripting in **python, Bash, and Powershell**. and worked with **P4V and Git** Version Controls and have basic knowledge in **MAKE** files, know how to prebuild systems with **cmake** and **premake**

Academic Achievements

Game-A-Thon , in VIT Chennai

Secured first place in the 'Game-a-thon,' a 6-hour game development competition organized by Qubit as an integral component of the technical fest Technovit at VIT Chennai. The project involved the creation of a 2D platformer game, with interesting player mechanics.

IEEE YESIST12 2023 , in NMIT Bengaluru

My Project "Design of IOT-Based Agricultural Drones for Effective Crop Management - 5G and Beyond" was **awarded as the BEST PROJECT** in the preliminaries of the International Project Innovation Challenge for students and young professionals IEEE YESIST12 2023-'Kaushalya' Open House Project Expo-2023 Organized by Department of ECE, Nitte Meenakshi Institute of Technology, Bengaluru in association with IEEE Bangalore Section on 4" May 2023.

ICCSST - 2023 , in Christ University Bengaluru

Presented a conference paper on the topic "Social Distancing and Face Mask Detection Using YOLO Object Detection Algorithm" at the International Conference on Computational Sciences and Sustainable Technologies jointly conducted by CHRIST (Deemed to be University), Bangalore, India, and Modern College of Business and Science, Muscat, Oman. on May 8th, 2023, Our paper is yet to be published (as of July 31st 2023)